

HisabDo Internship Bootcamp

Software Quality Assurance (SQA) Track

Day 6 – Authentication API Testing Using Postman

(API Test Cases Report)

Submitted By:

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Internship Duration:

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Organization:

HisabDo

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Test Case 1:

Scenario: Verify that a new user can successfully register with valid credentials.

Attribute	Description
Test Case ID	TC-01
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "testuser15", "email": "testuser15@example.com", "password": "Password@123" }
Expected Result	The API should successfully register a new user and return 201 Created with the user's details and a success message.
Actual Result	The API returned 201 Created with the registered user's details and the message "Users registered successfully and verification email has been sent on your email."
Status	Pass

Test Case 2:

Scenario: Verify that the API returns a validation error when the password field is missing during user registration.

Attribute	Description
Test Case ID	TC-02
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "testuser2", "email": "testuser2@example.com" }
Expected Result	The API should return 422 Unprocessable Entity with a validation error indicating that the password field is required.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and the validation error "Password is required".
Status	Pass

Test Case 3:

Scenario: Verify that the API returns validation errors when the email field is missing during user registration.

Attribute	Description
Test Case ID	TC-03
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "testuser3", "password": "Password@123" }
Expected Result	The API should return 422 Unprocessable Entity with validation errors indicating that the email field is required and invalid.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and the validation errors "Email is required" and "Email is invalid".
Status	Pass

Test Case 4:

Scenario: Verify that the API returns validation errors when an empty request body is submitted during user registration.

Attribute	Description
Test Case ID	TC-04
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json {}
Expected Result	The API should return 422 Unprocessable Entity with validation errors indicating that the required fields (email, username and password) are missing.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and validation errors for the missing email, username and password fields.
Status	Pass

Test Case 5:

Scenario: Verify that the API returns an error when an invalid email is used during user registration.

Attribute	Description
Test Case ID	TC-05
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "testuser5", "email": "invalidemail", "password": "Password@123" }
Expected Result	The API should return 422 Unprocessable Entity with a validation error indicating that the email format is invalid.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and the validation error "Email is invalid".
Status	Pass

Test Case 6:

Scenario: Verify that the API prevents registration when the email or username already exists.

Attribute	Description
Test Case ID	TC-06
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "testuser1", "email": "testuser1@example.com", "password": "Password@123" }
Expected Result	The API should return 409 Conflict and reject the registration because the email or username already exists.
Actual Result	The API returned 409 Conflict with the message "User with email or username already exists".
Status	Pass

Test Case 7:

Scenario: Verify that the API rejects registration when an excessively long email address is provided.

Attribute	Description
Test Case ID	TC-07
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	Valid username and password with an email address exceeding the allowed length.
Expected Result	The API should return 422 Unprocessable Entity with a validation error indicating that the email is invalid.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and the validation error "Email is invalid".
Status	Pass

Test Case 8:

Scenario: Verify that the API allows registration when the username contains special characters.

Attribute	Description
Test Case ID	TC-08
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "test11@user#\$%", "email": "testuser11@example.com", "password": "Password@123" }
Expected Result	The API should successfully register the user if special characters are permitted in the username. The response should return 201 Created with the registered user's details.
Actual Result	The API returned 201 Created and successfully registered the user with the username "test11@user#\$%".
Status	Pass

Test Case 9:

Scenario: Verify that a registered user can successfully log in using valid credentials.

Attribute	Description
Test Case ID	TC-09
API/Endpoint	https://api.freeapi.app/api/v1/users/login
Method	POST
Request	json { "email": "testuser11@example.com", "password": "Password@123" }
Expected Result	The API should authenticate the user and return 200 OK with the user's details, an access token, a refresh token and a success message.
Actual Result	The API returned 200 OK with the user's information, an access token, a refresh token and the message "User logged in successfully".
Status	Pass

Test Case 10:

Scenario: Verify that the API returns a validation error when an empty password is provided during login.

Attribute	Description
Test Case ID	TC-10
API/Endpoint	https://api.freeapi.app/api/v1/users/login
Method	POST
Request	json { "email": "testuser11@example.com", "password": "" }
Expected Result	The API should reject the login request and return 422 Unprocessable Entity with a validation error indicating that the password is required.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and the validation error "Password is required".
Status	Pass

Test Case 11:

Scenario: Verify that the API rejects login when an empty email is provided.

Attribute	Description
Test Case ID	TC-11
API/Endpoint	https://api.freeapi.app/api/v1/users/login
Method	POST
Request	json { "email": "", "password": "Password@123" }
Expected Result	The API should reject the login request and return 422 Unprocessable Entity with an email validation error.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and the validation error "Email is invalid".
Status	Pass

Test Case 12:

Scenario: Verify that the API rejects login when an invalid email format is provided.

Attribute	Description
Test Case ID	TC-12
API/Endpoint	https://api.freeapi.app/api/v1/users/login
Method	POST
Request	json { "email": "invalidemail", "password": "Password@123" }
Expected Result	The API should reject the login request and return 422 Unprocessable Entity with an email validation error.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and the validation error "Email is invalid".
Status	Pass

Test Case 13:

Scenario: Verify that the API rejects login when an incorrect password is provided.

Attribute	Description
Test Case ID	TC-13
API/Endpoint	https://api.freeapi.app/api/v1/users/login
Method	POST
Request	json { "email": "testuser1@example.com", "password": "WrongPassword@123" }
Expected Result	The API should reject authentication and return 401 Unauthorized with an error message indicating invalid credentials.
Actual Result	The API returned 401 Unauthorized with the message "Invalid user credentials" and no access token was generated.
Status	Pass

Test Case 14:

Scenario: Verify that the API returns validation errors when empty email and password fields are submitted during login.

Attribute	Description
Test Case ID	TC-14
API/Endpoint	https://api.freeapi.app/api/v1/users/login
Method	POST
Request	json { "email": "", "password": "" }
Expected Result	The API should reject the login request and return 422 Unprocessable Entity with validation errors for invalid email and missing password.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and validation errors "Email is invalid" and "Password is required".
Status	Pass

Test Case 15:

Scenario: Verify that the API safely handles SQL injection input during login and does not authenticate the request.

Attribute	Description
Test Case ID	TC-15
API/Endpoint	https://api.freeapi.app/api/v1/users/login
Method	POST
Request	json { "email": "" OR '1'='1", "password": "Password@123" }
Expected Result	The API should reject the malicious input and should not generate an authentication token.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and did not generate any token.
Status	Pass

Test Case 16:

Scenario: Verify that the API rejects XSS payloads during login and prevents malicious input from being processed.

Attribute	Description
Test Case ID	TC-16
API/Endpoint	https://api.freeapi.app/api/v1/users/login
Method	POST
Request	json { "email": "<script>alert('XSS')</script>", "password": "Password@123" }
Expected Result	The API should reject the malicious input, return a validation error and should not generate an authentication token.
Actual Result	The API returned 422 Unprocessable Entity with the message "Received data is not valid" and the validation error "Email is invalid".
Status	Pass

Test Case 17:

Scenario: Verify API behavior when an unsupported HTTP method is used.

Attribute	Description
Test Case ID	TC-17
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	GET
Request	No request body
Expected Result	The API should reject unsupported HTTP methods and return an appropriate error response such as 405 Method Not Allowed.
Actual Result	The API returned 200 OK and displayed the Swagger documentation HTML page instead of processing the API request.
Status	Pass

Test Case 18:

Scenario: Verify API behavior when an invalid Content-Type header is sent.

Attribute	Description
Test Case ID	TC-18
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "testcontent", "email": "testcontent@example.com", "password": "Password@123" }
Expected Result	The API should reject the request or handle the unsupported content type properly.
Actual Result	The API returned 422 Unprocessable Entity with validation errors indicating that email, username and password were not received.
Status	Pass

Test Case 19:

Scenario: Verify that the API rejects registration when the username length is below the minimum allowed limit.

Attribute	Description
Test Case ID	TC-19
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "ab", "email": "testuser12@example.com", "password": "Password@123" }
Expected Result	The API should reject usernames shorter than the minimum allowed length (3 characters).
Actual Result	The API returned 422 Unprocessable Entity with the message "Username must be at least 3 characters long".
Status	Pass

Test Case 20:

Scenario: Verify that the API accepts a username exceeding the normal length boundary during registration.

Attribute	Description
Test Case ID	TC-20
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "testusertestusertestusertestuser...", "email": "testuser17@example.com", "password": "Password@123" }
Expected Result	The API should validate the username length and either accept or reject the request according to the defined maximum length limit.
Actual Result	The API returned 201 Created and successfully registered the user with the long username.
Status	Pass

Test Case 21:

Scenario: Verify that the API validates passwords with the minimum allowed length during registration.

Attribute	Description
Test Case ID	TC-21
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "testuser20", "email": "testuser20@example.com", "password": "a" }
Expected Result	The API should reject passwords below the minimum required length.
Actual Result	The API returned 201 Created and successfully registered the user with a one-character password.
Status	Pass

Test Case 22:

Scenario: Verify that the API handles passwords exceeding the normal length boundary during registration.

Attribute	Description
Test Case ID	TC-22
API/Endpoint	https://api.freeapi.app/api/v1/users/register
Method	POST
Request	json { "username": "testuser22", "email": "testuser22@example.com", "password": "Password1234567890Password1234567890Password1234567890Password1234567890" }
Expected Result	The API should validate the password length and either accept or reject passwords according to the defined maximum length limit.
Actual Result	The API returned 201 Created and successfully registered the user with a long password.
Status	Pass

Test Case 23:

Scenario: Verify that an authenticated user can successfully logout and the active session is terminated.

Attribute	Description
Test Case ID	TC-23
API/Endpoint	https://api.freeapi.app/api/v1/users/logout
Method	POST
Request	Valid access token of logged-in user
Expected Result	The API should successfully logout the user, revoke the active session, and remove authentication tokens.
Actual Result	The API returned 200 OK with message "User logged out" and an empty data object.
Status	Pass

Test Case 24:

Scenario: Verify that a user cannot logout without providing an authentication token.

Attribute	Description
Test Case ID	TC-24
API/Endpoint	https://api.freeapi.app/api/v1/users/logout
Method	POST
Request	Logout request without Authorization header
Expected Result	API should reject the request and return an unauthorized error.
Actual Result	The API returned 401 Unauthorized with message "Unauthorized request".
Status	Pass

Test Case 25:

Scenario: Verify that the API rejects logout requests when an invalid authentication token is provided.

Attribute	Description
Test Case ID	TC-25
API/Endpoint	https://api.freeapi.app/api/v1/users/logout
Method	POST
Request	Invalid access token in Authorization header
Expected Result	API should reject the request and return an unauthorized error.
Actual Result	The API returned 401 Unauthorized and rejected the invalid token.
Status	Pass

Test Case 26:

Scenario: Verify that the API rejects logout requests when the access token is expired.

Attribute	Description
Test Case ID	TC-26
API/Endpoint	https://api.freeapi.app/api/v1/users/logout
Method	POST
Request	Expired access token in Authorization header
Expected Result	API should reject the request and return an unauthorized error.
Actual Result	The API returned 401 Unauthorized with message "jwt malformed" and rejected the token.
Status	Pass

Test Case 27:

Scenario: Verify that a logged-in user can successfully request a resend of the email verification mail.

Attribute	Description
Test Case ID	TC-27
API/Endpoint	https://api.freeapi.app/api/v1/users/resend-email-verification
Method	POST
Request	Valid access token of logged-in user
Expected Result	API should resend the verification email successfully.
Actual Result	The API returned 200 OK with message "Mail has been sent to your mail ID" and empty response data.
Status	Pass

Test Case 28:

Scenario: Verify that a logged-in user can successfully change their password by providing the correct old password and a valid new password.

Attribute	Description
Test Case ID	TC-28
API/Endpoint	https://api.freeapi.app/api/v1/users/change-password
Method	POST
Request	Valid access token, correct old password and a valid new password
Expected Result	The API should successfully change the user's password.
Actual Result	The API returned 200 OK with the message "Password changed successfully" and an empty data object.
Status	Pass

Test Case 29:

Scenario: Verify that the API rejects the password change request when an incorrect old password is provided.

Attribute	Description
Test Case ID	TC-29
API/Endpoint	https://api.freeapi.app/api/v1/users/change-password
Method	POST
Request	{ "oldPassword": "WrongPassword@123", "newPassword": "Password@456" }
Expected Result	API should return 400 Bad Request with the message "Invalid old password".
Actual Result	API returned 400 Bad Request with the message "Invalid old password".
Status	Pass

Test Case 30:

Scenario: Verify that the API successfully retrieves products sorted in descending order.

Attribute	Description
Test Case ID	TC-30
API/Endpoint	/products?sort=desc
Method	GET
Request	GET https://fakestoreapi.com/products?sort=desc
Expected Result	The API should return HTTP Status Code 200 OK and a non-empty JSON array containing products sorted in descending order.
Actual Result	The API returned HTTP Status Code 200 OK with a non-empty JSON array of products. All Postman assertions (Status code is 200, Response is an array and Response is not empty) passed successfully.
Status	Pass

Test Execution Summary

Metric	Count
Total Test Cases Designed	30
Total Test Cases Executed	30
Positive Test Cases	9
Negative Test Cases	18
Boundary Test Cases	3
Passed	30
Failed	0
Blocked	0
Pass Rate	100%